

Batch: A2 Roll No.: 16010324044

Experiment / assignment / tutorial No. 5

Grade: AA / AB / BB / BC / CC / CD / DD

Signature of the Staff In-charge with date

TITLE: VLAN concept using Cisco Packet Tracer and hardware.

AIM: To configure given network topology of VLAN using Cisco Packet Tracer.

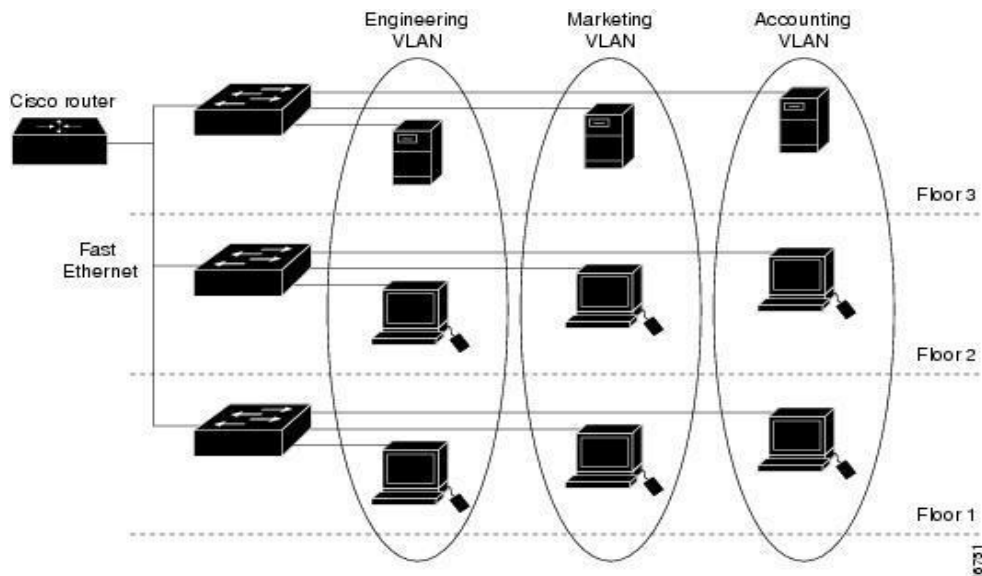
OUTCOME: Student will be able to understand layered architecture of communication network and physical layer services and perceive the significance of data link layer services and protocols.

Theory:

A VLAN is a group of devices on one or more LANs that are configured to communicate as if they were attached to the same wire, when in fact they are located on a number of different LAN segments. Because VLANs are based on logical instead of physical connections, they are extremely flexible.

VLANs define broadcast domains in a Layer 2 network. A broadcast domain is the set of all devices that will receive broadcast frames originating from any device within the set. Broadcast domains are typically bounded by routers because routers do not forward broadcast frames. Layer 2 switches create broadcast domains based on the configuration of the switch. Switches are multiport bridges that allow you to create multiple broadcast domains. Each broadcast domain is like a distinct virtual bridge within a switch.

One or many virtual bridges can be defined within a switch. Each virtual bridge, created in the switch, defines a new broadcast domain (VLAN). Traffic cannot pass directly to another VLAN (between broadcast domains) within the switch or between two switches. To interconnect two different VLANs, you must use routers or Layer 3 switch.



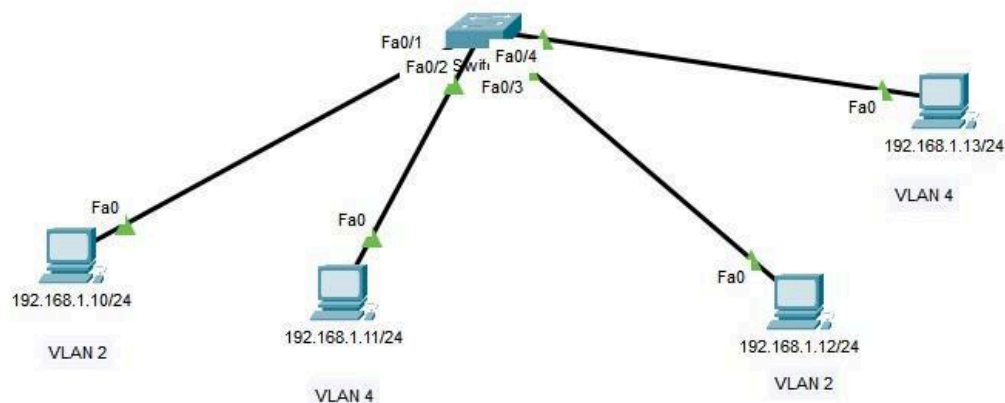
VLANs are often associated with IP subnetworks. For example, all of the end stations in a particular IP subnet belong to the same VLAN. Traffic between VLANs must be routed. You must assign LAN interface VLAN membership on an interface-by-interface basis (this is known as interface-based or static VLAN membership).

You can set the following parameters when you create a VLAN in the management domain:

- VLAN number
- VLAN name
- VLAN type
- VLAN state (active or suspended)
- Maximum transmission unit (MTU) for the VLAN
- Security Association Identifier (SAID)
- VLAN number to use when translating from one VLAN type to another

Procedure:

Part1:



1. To begin with the experiment, open Cisco Packet Tracer
2. Create the lab and configure the basic IP addressing for PCs, switch
3. Create vlan 2 and vlan 4 in the switch

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4. Assign interfaces to respective vlans 2 and 4
5. Give access to the ports as per the VLAN communication requirement.
6. ping from 192.168.1.10 to 192.168.1.12 and check the result
7. ping from 192.168.1.10 to 192.168.1.13 and check the result
8. Check the connectivity and comment in the conclusion

Output:

```
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\CELAB>ping 172.17.23.76

Pinging 172.17.23.76 with 32 bytes of data:
Reply from 172.17.23.76: bytes=32 time<1ms TTL=128
Reply from 172.17.23.76: bytes=32 time=1ms TTL=128
Reply from 172.17.23.76: bytes=32 time=1ms TTL=128
Reply from 172.17.23.76: bytes=32 time=1ms TTL=128

Ping statistics for 172.17.23.76:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\CELAB>ping 172.17.23.77

Pinging 172.17.23.77 with 32 bytes of data:
Reply from 172.17.23.77: bytes=32 time=1ms TTL=128
Reply from 172.17.23.77: bytes=32 time=1ms TTL=128
Reply from 172.17.23.77: bytes=32 time=1ms TTL=128
Reply from 172.17.23.77: bytes=32 time<1ms TTL=128

Ping statistics for 172.17.23.77:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\CELAB>ping 172.17.23.78

Pinging 172.17.23.78 with 32 bytes of data:
Reply from 172.17.23.78: bytes=32 time=1ms TTL=128
Reply from 172.17.23.78: bytes=32 time=1ms TTL=128
Reply from 172.17.23.78: bytes=32 time=1ms TTL=128
Reply from 172.17.23.78: bytes=32 time<1ms TTL=128

Ping statistics for 172.17.23.78:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\CELAB>
```

```
C:\Users\CELAB>ping 172.17.23.76

Pinging 172.17.23.76 with 32 bytes of data:
Reply from 172.17.23.76: bytes=32 time=1ms TTL=128
Reply from 172.17.23.76: bytes=32 time<1ms TTL=128
Reply from 172.17.23.76: bytes=32 time=1ms TTL=128
Reply from 172.17.23.76: bytes=32 time<1ms TTL=128

Ping statistics for 172.17.23.76:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\CELAB>ping 172.17.23.77

Pinging 172.17.23.77 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 172.17.23.75: Destination host unreachable.
Reply from 172.17.23.75: Destination host unreachable.

Ping statistics for 172.17.23.77:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),

C:\Users\CELAB>ping 172.17.23.78

Pinging 172.17.23.78 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 172.17.23.75: Destination host unreachable.
Reply from 172.17.23.75: Destination host unreachable.

Ping statistics for 172.17.23.78:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),

C:\Users\CELAB>
```

```
e
Rack5_SW3#show vlan
```

VLAN	Name	Status	Ports
1	default	active	Gi0/1, Gi0/2, Gi0/3, Gi0/4 Gi0/9, Gi0/10, Gi0/11, Gi0/12
2	VLAN0002	active	
3	VLAN0003	active	
4	VLAN0004	active	
5	VLAN0005	active	
6	VLAN0006	active	
7	VLAN0007	active	
8	VLAN0008	active	
9	SSK	active	
10	HR	active	
11	mech	active	
12	it	active	
13	comp	active	
14	extc	active	
15	VLAN0015	active	
16	VLAN0016	active	
17	VLAN0017	active	
18	VLAN0018	active	
20	Accounts	active	
21	a	active	
22	b	active	
23	VLAN0023	active	
24	VLAN0024	active	
25	hrk	active	
26	exp	active	
27	hetin	active	
28	om	active	
30	kjsce	active	
31	kjsse	active	
32	akaash	active	Gi0/5, Gi0/6
33	hi	active	Gi0/7, Gi0/8
76	Harshvardhan	active	
78	Raj	active	
99	Management	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	

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--More--
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Part 2:

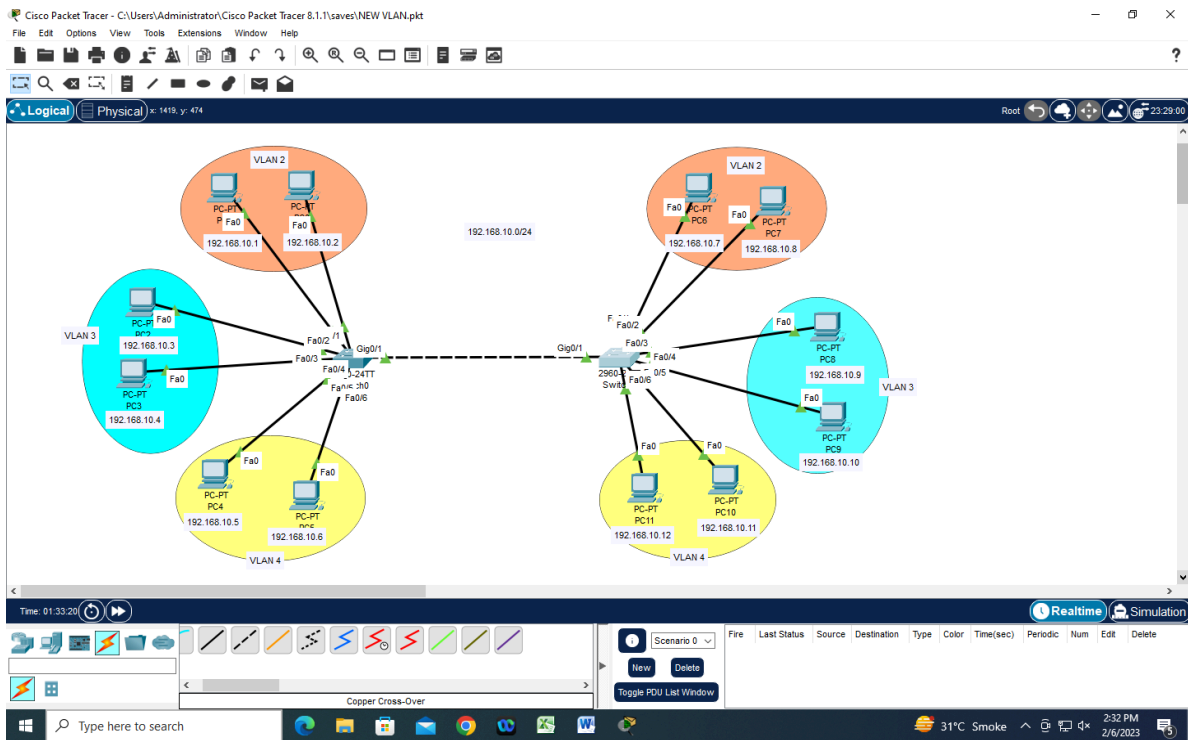
1. To begin with the experiment, open Cisco Packet Tracer
2. Create the lab and configure the basic IP addressing for PCs, switch
3. Using the command line interface of the router or switch enter the global configuration mode of router using command **configure terminal**
4. Configure the VLAN as per the network diagram
5. Assign interfaces to respective vlans 2 ,3 and 4 using command
S1(config)# **switchport mode access**
S1(config)# **switchport access vlan 2**
6. Give access to each vlan connected to the switch
7. Configure the gigabit Ethernet port for the switch in trunk mode using command
S1(config)# **switchport mode trunk**

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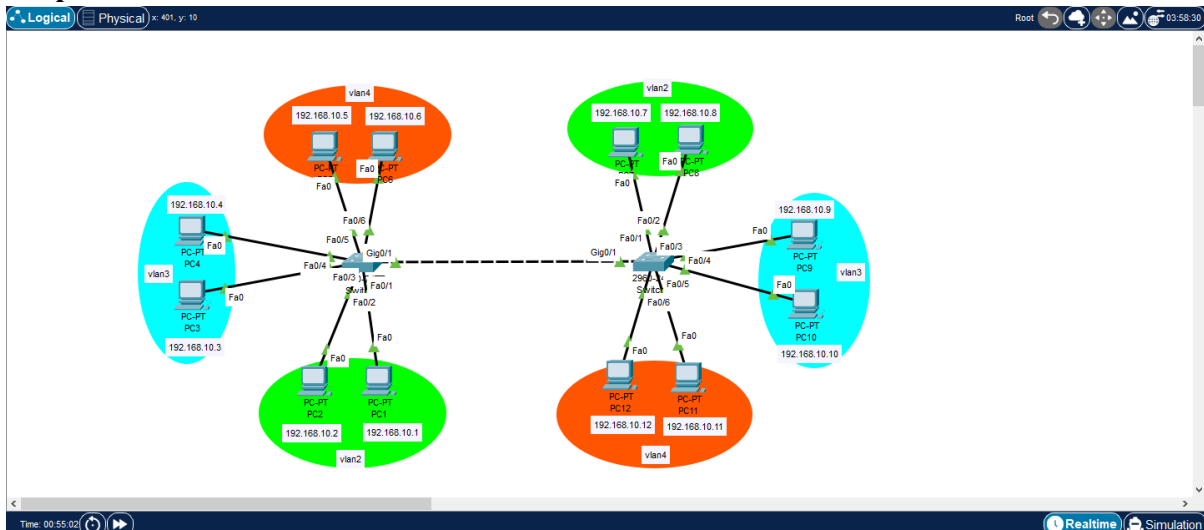
S1(config)# **switchport trunk allowed vlan 2,3,4**

8. Using **ping** command, test the ability of the source computer to each device in the network.

Example:



Output:



```
C:\>ping 192.168.10.6

Pinging 192.168.10.6 with 32 bytes of data:

Reply from 192.168.10.6: bytes=32 time<1ms TTL=128
Reply from 192.168.10.6: bytes=32 time=1ms TTL=128
Reply from 192.168.10.6: bytes=32 time=1ms TTL=128
Reply from 192.168.10.6: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
C:\>ping 192.168.10.8

Pinging 192.168.10.8 with 32 bytes of data:

Reply from 192.168.10.8: bytes=32 time<1ms TTL=128
Reply from 192.168.10.8: bytes=32 time<1ms TTL=128
Reply from 192.168.10.8: bytes=32 time<1ms TTL=128
Reply from 192.168.10.8: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
C:\>ping 192.168.10.6

Pinging 192.168.10.6 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.6:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.10.8

Pinging 192.168.10.8 with 32 bytes of data:

Reply from 192.168.10.8: bytes=32 time=14ms TTL=128
Reply from 192.168.10.8: bytes=32 time<1ms TTL=128
Reply from 192.168.10.8: bytes=32 time<1ms TTL=128
Reply from 192.168.10.8: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 14ms, Average = 3ms

C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

CONCLUSION

In this experiment, I successfully configured the VLAN network using Cisco Packet Tracer. VLANs 2, 3, and 4 were created and the respective switch ports were assigned to them. Access ports and trunk ports were configured as required, with the trunk allowing VLANs 2, 3, and 4. Connectivity was successfully verified using the ping command. Devices belonging to the same VLAN were able to communicate, while communication between different VLANs was restricted without inter-VLAN routing. Thus, the experiment demonstrated the concept of VLANs, broadcast domains, access ports, and trunking.

Signature of faculty in-charge

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